Notes: Akcigit, Hanley, and Stantcheva (Econometrica, 2022)

Optimal Taxation and R&D Policies

Contents

1 Big picture

- **Question.** How should a government design corporate taxes and R&D subsidies when firms create innovation spillovers, but firms' research productivity is privately known and some innovation inputs are unobservable?
- **Setting.** Firms are heterogeneous innovators. They choose observable R&D investment and unobservable R&D effort to improve product quality over time. Innovation spills over to other firms through lower production costs, and patent protection gives innovators monopoly power.
- **Core challenge.** With full information, the government could use Pigouvian subsidies and prizes to correct spillovers and nonappropriability. With asymmetric information, however, subsidizing R&D too aggressively can let high-productivity firms mimic low-productivity firms and extract rents.
- **Main theoretical result.** The optimal policy can be expressed through two wedges: a *corporate/profit wedge* on the unobservable effort margin and an *R&D wedge* on the observable investment margin. Each wedge combines three forces: a Pigouvian spillover correction, a monopoly-distortion correction, and a screening term created by private information.
- **Key mechanism.** The sign and size of the screening term for R&D depend on the *relative complementarity* of observable R&D with unobservable effort versus with private research productivity. If R&D is more complementary to firm type than to effort, subsidizing R&D creates large information rents and the planner dampens subsidies for lower-productivity firms.
- **Empirical strategy.** The paper estimates the model using firm-level Census data matched to USPTO patent data. Patent citations proxy for innovation quality, reported R&D spending maps to observable R&D investment, and firm sales and profits discipline production-side objects.
- **Main quantitative findings.**
  - R&D is highly complementary to firm research productivity: the estimated substitution parameter is $\rho_{\theta r} = 1.88$.
  - Research productivity is moderately persistent: $\tilde{p} = 0.63$.
  - Spillovers are economically meaningful: the estimated production externality is $\zeta = 0.018$.
- **Main policy result.** Very simple policies can do nearly as well as the unrestricted optimum. A linear corporate tax combined with a nonlinear R&D subsidy achieves about **97.3%** of the welfare gain from the full optimal mechanism. By contrast, a policy close to the current US system achieves only about **18%** of that gain.
- **Economic takeaway.** The paper argues that good innovation policy is not "subsidize all R&D as much as possible." Instead, the optimal policy must screen firms, target spillovers, and decide whether it is better to reward *observable spending* or *realized performance*. In the data, rewarding performance through lower profit wedges for high-productivity firms is central.

2 Model, data, and key objects

2.1 Firms, product quality, and innovation inputs

- Firms produce differentiated intermediate goods and improve their product quality through innovation.
- Product quality evolves according to

$$q_t = H(q_{t-1}, \lambda_t),$$

where λ_t is the quality improvement or *step size*.

- The step size depends on three ingredients:

$$\lambda_t = \lambda_t(r_{t-1}, l_t, \theta_t),$$

where

  - r_{t-1} is **observable R&D investment** (scientists' pay, labs, materials, equipment),
  - l_t is **unobservable R&D effort** (managerial input, organizational effort, nonverifiable research effort), and
  - θ_t is the firm's **research productivity type**.

- R&D investment has monetary cost $M_t(r_t)$, with $M_t'(r_t) > 0$ and $M_t''(r_t) \geq 0$.
- R&D effort has private cost $\phi_t(l_t)$, increasing and convex.
- The type θ_t is the firm's private research productivity: management quality, organizational efficiency, idea quality, and other characteristics that determine how effectively the firm turns inputs into innovation.
- The type follows a stochastic Markov process $f^t(\theta_t \mid \theta_{t-1})$. The firm has information about its current type and some persistence in future type, but does not perfectly know its future innovation success when it chooses current R&D spending.

2.2 Spillovers, production, and market structure

- Innovation creates **quality spillovers**. Aggregate quality is

$$\bar{q}_t = \int q_t(\theta^t) P(\theta^t)\, d\theta^t.$$

- Higher aggregate quality lowers each firm's production cost $C_t(k, \bar{q}_t)$. This is the "building on the shoulders of giants" channel: one firm's innovations improve production possibilities for others.
- The final good is produced competitively from intermediate goods. Intermediate goods firms, however, have patent protection and monopoly power in the benchmark environment.
- Given quality, a monopolist chooses quantity to maximize production profits gross of R&D costs:

$$\pi(q_t(\theta^t), \bar{q}_t) = \max_k \{p(q_t(\theta^t), k)k - C(k, \bar{q}_t)\}.$$

- The paper studies the benchmark case in which the government *cannot* condition policy on quantity, so it must take the private market between intermediate- and final-goods producers as given.

2.3 Welfare objects and the first-best benchmark

- Let firm payoff in period t be

$$v_t(\theta^t) = T_t(\theta^t) - \phi_t(l_t(\theta^t)),$$

  where T_t is the transfer net of production and R&D costs.

- Social welfare is consumer surplus plus firm surplus, net of informational rents. The benchmark contract-theory case sets the planner's objective to virtual surplus ($\chi = 1$ in the paper).

- With full information and nondistortionary transfers, the planner solves a first-best problem. When quality law of motion is simplified to

$$q_t = (1-\delta)q_{t-1} + \lambda_t,$$

  the first-best conditions for R&D and effort become

$$M_t'(r_t(\theta^t)) = \frac{1}{R} E\left[\sum_{s=t+1}^{\infty}\left(\frac{1-\delta}{R}\right)^{s-t-1}\left(\frac{\partial \widetilde{Y}_s^*}{\partial q_s} + \frac{\partial \widetilde{Y}_s^*}{\partial \bar{q}_s}\right)\frac{\partial \lambda_{t+1}}{\partial r_t}\right],$$
$$\phi_t'(l_t(\theta^t)) = E\left[\sum_{s=t}^{\infty}\left(\frac{1-\delta}{R}\right)^{s-t}\left(\frac{\partial \widetilde{Y}_s^*}{\partial q_s} + \frac{\partial \widetilde{Y}_s^*}{\partial \bar{q}_s}\right)\frac{\partial \lambda_t}{\partial l_t}\right].$$

- *Interpretation.* In the first best, the planner fully internalizes both private gains from quality and spillovers through aggregate quality.

2.4 Information structure and why it matters

- The core private information is the history of research productivities θ^t and unobservable R&D effort l_t.
- The government observes:
  - observable R&D spending r_t,
  - the innovation step size λ_t,
  - realized quality q_t, and
  - in the benchmark case, not quantity k_t.
- The paper interprets patent counts and forward citations as an empirical proxy for observed innovation outcomes.
- Why asymmetric information is important:
  - venture capital exists precisely because screening innovators is hard,
  - insider gains are larger at R&D-intensive firms,

- reported R&D can be relabeled or mismeasured,
- and in the paper's own data, even very rich observables predict innovation quality poorly.

- Specifically, regressions of citations-weighted patents on a large set of observables yield an R^2 that barely exceeds **0.3**. This is suggestive evidence that governments cannot simply "tag" firm quality using observables.

2.5 Data, empirical mapping, and key moments

- **Benchmark data.** Census Bureau Longitudinal Business Database and Census of Manufacturers, matched to USPTO patent data from the NBER patent database.
- **Alternative sample.** The supplement also reports results using COMPUSTAT matched to patents.
- **Model-data mapping.**
  - $M(r)$ corresponds to reported R&D expenditure.
  - The step size λ_t is measured by forward citations received by a firm's patents filed in year t.
  - Quality is the depreciation-adjusted stock of citation-weighted patent quality:

$$q_t = (1-\delta)q_{t-1} + \lambda_t.$$

  - Sales and profits are observed directly in the firm data.
- **Status quo policy in the estimation.** The benchmark "current US policy" is approximated as a 23% effective corporate tax rate and a 19% effective R&D subsidy rate.

3 Models and empirical specifications (all equations + interpretation)

3.1 Innovation technology and Hicksian complementarity

The paper's central production object is the step size $\lambda_t(r_{t-1}, l_t, \theta_t)$. To measure how two inputs interact in producing innovation, the paper uses the Hicksian coefficient of complementarity. For any two variables $(x, y) \in \{r, l, \theta\}^2$,

$$\rho_{xy} = \frac{\partial^2 \lambda}{\partial x\, \partial y} \frac{\lambda}{\lambda_x \lambda_y}.$$

- If $\rho_{xy} > 0$, the two inputs are complementary.
- If $\rho_{xy} = 0$, they are additively separable.
- If ρ_{xy} is larger, screening becomes harder when the observable input disproportionately helps high-type firms.

This object is crucial because the sign of the screening term in the optimal R&D subsidy depends on

$$\rho_{lr} - \rho_{\theta r}.$$

Interpretation.

- If observable R&D is more complementary to *effort* than to *type*, subsidizing R&D helps draw out unobservable effort and the planner wants a larger subsidy.
- If observable R&D is more complementary to *type* than to *effort*, subsidizing R&D mainly hands rents to high-productivity firms and the planner dampens subsidies for lower-productivity firms.

3.2 Firm problem in laissez-faire

Absent government intervention, a firm of initial type θ_1, initial quality q_0, and initial R&D stock r_0 chooses quality, quantity, R&D investment, and effort to maximize

$$\sum_{t=1}^{\infty}\left(\frac{1}{R}\right)^{t-1}\int_{\Theta^t}\Big[p(q_t(\theta^t),k_t(\theta^t))k_t(\theta^t)-C(k_t(\theta^t),\bar{q}_t)-M_t(r_t(\theta^t))-\phi_t(l_t(\theta^t))\Big]P(\theta^t\mid\theta_1)\,d\theta^t.$$

Interpretation.

- The firm internalizes its own production profits from higher quality.
- It does *not* internalize the positive effect of its innovation on other firms through $\bar{q}_t$.
- It also values innovation through monopoly profits under patent protection, not through total social value.

3.3 Direct revelation mechanism and continuation utility

To characterize constrained efficiency, the paper lets the government run a direct revelation mechanism. In each period, the firm reports a type history $\hat{\theta}^t(\theta^t)$, and the planner assigns step sizes, R&D investments, and transfers as functions of reported histories.

Let σ denote a reporting strategy. The continuation value after history θ^t under σ is

$$V^\sigma(\theta^t)=T_t(\hat{\theta}^t(\theta^t))-\phi_t\Big(l_t(\lambda_t(\hat{\theta}^t(\theta^t)),r(\hat{\theta}^{t-1}(\theta^{t-1})),\theta_t)\Big)+\frac{1}{R}\int V^\sigma(\theta^{t+1})f^{t+1}(\theta_{t+1}\mid\theta_t)\,d\theta_{t+1}.$$

Truth-telling requires

$$V(\theta^t)\geq V^\sigma(\theta^t)\qquad\text{for all reporting strategies }\sigma.$$

Interpretation. The planner does not choose a tax schedule directly at first. It chooses allocations subject only to incentive compatibility, and later shows how taxes can implement those allocations.

3.4 Envelope representation and planner's problem

The paper uses a first-order approach. Let $I_{1t}(\theta^t)$ denote the impulse response of period-t type to a shock in period 1. For an AR(1) process, this collapses to

$$I_{1t}(\theta^t)=\tilde{p}^{\,t-1}.$$

The derivative-form envelope condition is

$$\frac{\partial V(\theta^t)}{\partial\theta_t}=E\left[\sum_{s=t}^{\infty}I_{ts}(\theta^s)\left(\frac{1}{R}\right)^{s-t}\frac{\partial v_s(\theta^s)}{\partial\theta_s}\,\middle|\,\theta_t\right].$$

The integral form implies that the planner must leave informational rents to induce truthful reporting.

The planner then solves the problem in two steps:

- an **inner / partial problem** taking the aggregate quality path $\bar{q} = (\bar{q}_1, \ldots, \bar{q}_T)$ as given and solving for optimal allocations,
- and an **outer / full problem** choosing the sequence $\bar{q}$ to satisfy the consistency condition

$$\int q_t(\theta^t) P(\theta^t)\, d\theta^t = \bar{q}_t.$$

Interpretation. This is how the paper incorporates spillovers between agents inside a dynamic mechanism design problem without solving a full many-firm game directly.

3.5 Wedges: definitions and economic meaning

The constrained-efficient allocation is easiest to understand through two wedges.

Corporate / profit wedge:

$$\tau(\theta^t) = E\left[\sum_{s=t}^{\infty} \left(\frac{1-\delta}{R}\right)^{s-t} \frac{\partial \pi_s(q_s(\theta^s), \bar{q}_s)}{\partial q_s(\theta^s)} \frac{\partial \lambda_t(\theta^t)}{\partial l_t(\theta^t)}\right] - \phi_t'(l_t(\theta^t)).$$

R&D investment wedge:

$$s(\theta^t) = M_t'(r_t(\theta^t)) - \frac{1}{R} E\left[\sum_{s=t+1}^{\infty} \left(\frac{1-\delta}{R}\right)^{s-t-1} \frac{\partial \pi_s(q_s(\theta^s), \bar{q}_s)}{\partial q_s(\theta^s)} \frac{\partial \lambda_{t+1}(\theta^{t+1})}{\partial r_t(\theta^t)}\right].$$

- A positive $\tau(\theta^t)$ means effort is distorted downward relative to laissez-faire.
- A positive $s(\theta^t)$ means the planner wants more R&D investment than laissez-faire, conditional on effort.
- The profit wedge behaves like an implicit corporate tax or profit subsidy on the unobservable effort margin.
- The R&D wedge behaves like an implicit subsidy on observable R&D spending.

3.6 Optimal wedge formulas and interpretation

The paper defines two useful objects:

$$\mathcal{J}_t(\theta^t) = \sum_{s=t}^{\infty} \left(\frac{1-\delta}{R}\right)^{s-t} \frac{\partial \pi_s(q_s(\theta^s), \bar{q}_s)}{\partial q_s(\theta^s)}, \qquad Q_t(\theta^t) = \sum_{s=t}^{\infty} \left(\frac{1-\delta}{R}\right)^{s-t} \frac{\partial \widetilde{Y}_s(q_s(\theta^s), \bar{q}_s)}{\partial q_s(\theta^s)}.$$

Here $\mathcal{J}_t$ is the private value of future quality through profits, while Q_t is the social value of future quality through net output.

Proposition 1 shows that the optimal wedges can be written as

$$
\begin{aligned}
\tau(\theta^t) = &-E\left[\sum_{s=t}^{\infty}\left(\frac{1-\delta}{R}\right)^{s-t}\eta_s\right]\frac{\partial\lambda_t}{\partial l_t} - \left(Q_t(\theta^t)-\mathcal{J}_t(\theta^t)\right)\frac{\partial\lambda_t}{\partial l_t} \\
&+\frac{1-F_1(\theta_1)}{f_1(\theta_1)}I_{1t}(\theta^t)\phi_t'\frac{\lambda_{\theta,t}}{\lambda_t}\left(\frac{1}{\varepsilon_{l,1-\tau,t}\,\varepsilon_{\lambda,l,t}}+\rho_{\theta l,t}\right),
\end{aligned}
$$

$$
\begin{aligned}
s(\theta^t) = &\frac{1}{R}E\left[\sum_{s=t+1}^{\infty}\left(\frac{1-\delta}{R}\right)^{s-t-1}\eta_s\frac{\partial\lambda_{t+1}}{\partial r_t}\right]+\frac{1}{R}E\left[\left(Q_{t+1}(\theta^{t+1})-\mathcal{J}_{t+1}(\theta^{t+1})\right)\frac{\partial\lambda_{t+1}}{\partial r_t}\right] \\
&+\frac{1}{R}E\left[\frac{1-F_1(\theta_1)}{f_1(\theta_1)}I_{1,t+1}(\theta^{t+1})\phi_{t+1}'\frac{\lambda_\theta\lambda_r}{\lambda_l\lambda}(\rho_{lr}-\rho_{\theta r})\right].
\end{aligned}
$$

The spillover multiplier is

$$
\eta_t = \int \frac{\partial\widetilde{Y}(q_t(\theta^t),\bar{q}_t)}{\partial\bar{q}_t}P(\theta^t)\,d\theta^t.
$$

Interpretation. Each wedge has three parts:

- **Pigouvian correction.** Firms do not internalize the spillover from higher aggregate quality, so this part pushes the planner to subsidize innovation inputs.
- **Monopoly quality valuation correction.** Firms value quality through monopoly profits, but the planner values quality through total social surplus. This also tends to raise incentives relative to laissez-faire.
- **Screening term.** Because high-productivity firms can mimic low-productivity firms, the planner must distort allocations to save on informational rents.

Two especially important comparative statics follow directly:

- More persistent or more dispersed private types make screening harder and increase the magnitude of distortions.
- If $\rho_{\theta r} > \rho_{lr}$, observable R&D is more complementary to high private type than to unobservable effort, so the planner dampens R&D incentives for lower-productivity firms.

3.7 Implementation through taxes and subsidies

The paper shows that the constrained-efficient allocation can be implemented by an age-dependent tax function that need not condition on the entire history. Under AR(1)-type persistence, a relatively parsimonious implementation is

$$
T_t(q_t, r_t, q_{t-1}, r_{t-1}, q_1),
$$

or equivalently, because profits are an immediate function of quality,

$$
\widetilde{T}_t(\pi_t, r_t, \pi_{t-1}, r_{t-1}, \pi_1).
$$

Interpretation.

- This is nontrivial. In many dynamic tax problems, optimal taxes condition on full histories.
- Here, past quality can summarize the relevant payoff consequences of past investment, allowing the planner to reduce history dependence to current outcomes, one lag, and the first period's quality.

3.8 Parameterization and estimation strategy

The paper uses the functional forms shown in Table I:

- consumer valuation: $Y(q_t, k_t) = \frac{1}{1-\beta} q_t^{\beta} k_t^{1-\beta}$,
- cost function: $C_t(k, \bar{q}_t) = k/\bar{q}_t^{\zeta}$,
- quality law of motion: $q_t = (1-\delta)q_{t-1} + \lambda_t$,
- step size:
$$\lambda_t(r_{t-1}, l_t, \theta_t) = \left(\alpha r_{t-1}^{1-\rho_{\theta r}} + (1-\alpha)\theta_t^{1-\rho_{\theta r}}\right)^{\frac{1}{1-\rho_{\theta r}}} l_t,$$
- effort cost: $\phi_t(l_t) = \kappa_l \, l_t^{1+\gamma}/(1+\gamma)$,
- R&D cost: $M_t(r_t) = \kappa_r \, r_t^{1+\eta}/(1+\eta)$,
- type process: $\log \theta_t = \tilde{p} \log \theta_{t-1} + (1-\tilde{p})\mu_\theta + \varepsilon_t$.

Externally calibrated parameters (Table II).

- $R = 1.05$, $\delta = 0.1$, $\beta = 0.15$, $\eta = 1.5$, $\mu_\theta = 0$, $r_0 = 1$, horizon $T = 30$.

Internally estimated parameters (Table II).

- $\alpha = 0.483$, $\rho_{\theta r} = 1.88$, $\sigma_\varepsilon = 0.320$, $\tilde{p} = 0.63$,
- $\kappa_l = 0.69$, $\kappa_r = 0.055$, $\gamma = 0.86$, initial support width $\mathcal{S}_1 = 1.91$,
- spillover parameter $\zeta = 0.018$.

The parameter vector is estimated by minimizing the distance between model moments and data moments in a status quo economy with the current US policy in place.

3.9 Policy classes used in the quantitative comparison

The paper compares the unrestricted mechanism with simpler policy classes:

- current US-style linear corporate tax and linear R&D subsidy,
- optimal linear tax plus linear subsidy,
- linear tax with an interaction term in R&D,
- Heathcote–Storesletten–Violante (HSV)-type nonlinear profit taxes and nonlinear R&D subsidies,
- and hybrid cases that linearize only one side of the system.

The central quantitative question is not whether the full mechanism is theoretically best — it is — but whether a much simpler policy can approximate it closely enough to be practically useful.

4 Identification and economic logic (what makes the results credible)

4.1 What asymmetric information changes relative to the first best

- With **full information**, the planner would simply correct the spillover with a Pigouvian subsidy and correct nonappropriability with a prize or equivalent policy.
- With **private information**, the planner must also leave rents to induce truthful revelation and sufficient effort. The screening term is therefore a genuine extra distortion, not just a reinterpretation of the Pigouvian logic.
- This is why the paper's central object is not just "the size of spillovers," but the interaction among spillovers, market power, and informational rents.

4.2 What identifies R&D-type complementarity

- The elasticity of patent quality with respect to R&D spending is the key moment disciplining how R&D translates into innovation.
- In the data, higher-productivity firms obtain disproportionately higher returns from a given R&D investment. This is what pushes the estimate to $\rho_{\theta r} = 1.88 > 1$.
- Economically, this tells us that observable R&D spending is especially valuable in the hands of already-high-productivity firms. That is exactly the situation in which broad subsidies create information rents and must be targeted carefully.

4.3 What identifies persistence and dispersion of private productivity

- Within-firm variation in patent quality helps identify the stochastic process for private type.
- Cross-firm variation in patent quality, separately for young and old firms, helps identify dispersion and life-cycle heterogeneity.
- The estimated persistence $\tilde{p} = 0.63$ is moderate: enough to make dynamic incentives matter, but not so high that screening distortions fail to decay with age.

4.4 What identifies spillovers and input costs

- The spillover parameter ζ is identified by indirect inference using evidence from Bloom, Schankerman, and Van Reenen (2013): the model is asked to match the effect of rivals' R&D on a firm's sales.
- The mean R&D-to-sales ratio disciplines the importance of R&D in the step size function.
- Sales growth disciplines the scale of R&D costs.
- The elasticity of R&D investment to user cost, targeted at -0.35, provides an extra check that the model's implied response to policy is in line with reduced-form evidence.

4.5 Why the patent-firm match matters

- The match between firm data and patent data gives the model separate empirical handles on
  - inputs into innovation (R&D spending),
  - innovation outcomes (patents and citations), and
  - downstream business performance (sales and profits).
- This is crucial because the theory distinguishes between quality improvements, profits, and social output. Without matched data, many of these objects would be impossible to discipline separately.

4.6 Why simpler policies can approximate the optimum

- The unrestricted mechanism is history dependent and nonlinear, but the estimated wedges display fairly systematic patterns.
- Most importantly, the planner wants a strongly nonlinear *R&D subsidy schedule*: lower marginal subsidies at higher R&D levels.
- By contrast, a fully nonlinear profit tax adds little beyond a simple linear profit tax, because low-profit firms do not pay much tax to begin with. So being too generous to them is not very costly in welfare terms.

4.7 Relation to the literature

- **Mechanism design / dynamic public finance.** The paper extends the Pavan-Segal-Toikka and Stantcheva approach to a dynamic firm problem with spillovers.
- **Innovation policy.** It brings asymmetric information to a literature that often focuses on spillovers but treats firm quality as observable.
- **Corporate taxation.** The framework is broader than innovation. In principle, it can be used to study firm taxation in settings with hidden productivity and dynamic investment choices more generally.

5 Tables and figures

5.1 Figure 1: model summary

Read:

- The figure is a clean map of the environment. It places the household at the top, the government in the middle, and intermediate-goods producers together with the final-goods producer below.
- The key visual message is that the government affects firms through R&D and tax policies, while intellectual property rights determine the demand environment facing intermediate-goods producers.

- The diagram also makes clear why spillovers matter: aggregate quality $\bar{q}_t$ feeds back into firms' production costs.
- This figure is useful because it reminds you that the paper is *not* just a reduced-form tax exercise. It is a structural general-equilibrium environment with market power, innovation, and spillovers.

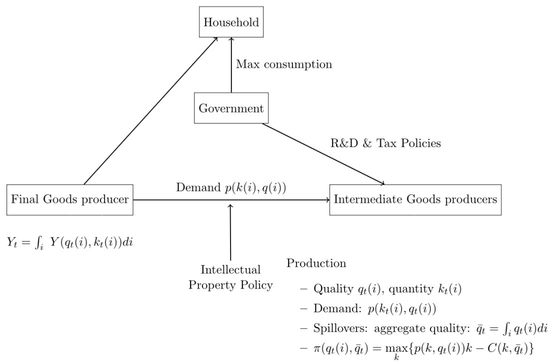

FIGURE 1.—Model summary.

TABLE I

FUNCTIONAL FORMS.

Function	Notation	Functional Form
Consumer valuation	$Y(q_t, k_t)$	$\frac{1}{1-\beta} q_t^{\beta} k_t^{1-\beta}$
Cost function	$C_t(k, \bar{q}_t)$	$\frac{k}{\bar{q}_t^{\zeta}}$
Quality accumulation	$H(q_{t-1}, \lambda_t)$	$q_t = (1-\delta)q_{t-1} + \lambda_t$
Step size	$\lambda_t(r_{t-1}, l_t, \theta_t)$	$(\alpha r_{t-1}^{1-\rho_{\theta r}} + (1-\alpha)\theta_t^{1-\rho_{\theta r}})^{\frac{1}{1-\rho_{\theta r}}} l_t$
Disutility of effort	$\phi_t(l_t)$	$\kappa_l \frac{l_t^{1+\gamma}}{1+\gamma}$
Cost of R&D	$M_t(r_t)$	$\kappa_r \frac{r_t^{1+\eta}}{1+\eta}$
Stochastic type process	$f^t(\theta_t \mid \theta_{t-1})$	$\log \theta_t = \tilde{p} \log \theta_{t-1} + (1-\tilde{p})\mu_\theta + \epsilon_t$
Distribution of heterogeneity θ_1	$f^1(\theta_1)$	$f^1(\theta_1) = \frac{I_{\Theta_1}(\theta_1)}{\theta_1[\underline{\theta}_1 - \bar{\theta}_1]}$
Initial quality level	q_0	0

Note: $I_{\Theta_1}(\theta_1)$ denotes the indicator function equal to 1 if θ_1 is in the set $\Theta_1 = [\underline{\theta}_1, \bar{\theta}_1]$.

5.2 Table I: functional forms

Read:

- Table I pins down the exact objects used in the quantitative work. The most important line is the step-size technology, which is CES in observable R&D and private type, and multiplicative in effort.
- That specification is why the single parameter $\rho_{\theta r}$ becomes so central: it governs whether observable R&D mostly amplifies hidden type or instead works through hidden effort.
- The remaining choices are deliberately standard: isoelastic cost functions, linear quality accumulation with depreciation, and a log AR(1)-style private type process.

5.3 Table II and Table III: calibrated parameters, estimates, and matched moments

TABLE II

PARAMETER VALUES.

Parameter	Symbol	Value	Standard Error
	External Calibration		
Interest rate	R	1.05	
Intangibles depreciation	δ	0.1	
Knowledge share	β	0.15	
R&D cost elasticity	η	1.5	
Level of types	μ_θ	0.00	
Initial R&D stock	r_0	1.0	
Program horizon	T	30	
	Internal Calibration		
R&D share	α	0.483	(0.025)
R&D-type substitution	$\rho_{\theta r}$	1.88	(0.126)
Type variance	σ_ϵ	0.320	(0.014)
Type persistence	$\tilde{p}$	0.63	(0.022)
Scale of disutility	κ_l	0.69	(0.050)
Scale of R&D cost	κ_r	0.055	(0.003)
Effort cost elasticity	γ	0.86	(0.052)
Support width for θ_1	Θ^1	1.91	(0.097)
Production externality	ζ	0.018	(0.001)

TABLE III

MOMENTS.

Moment	Target	Simulation	Standard Error
M1. Patent quality-R&D elasticity	0.88	0.97	(0.0009)
M2. R&D/Sales mean	0.041	0.035	(0.0025)
M3. Sales growth (DHS) mean	0.06	0.07	(0.005)
M4. Within-firm patent quality coeff of var	0.63	0.76	(0.0017)
Across-firm patent quality coeff of var:			
M5. Young firms	1.06	1.05	(0.0012)
M6. Older firms	0.99	0.81	(0.0016)
M7. Patent quality young/old	1.04	1.08	(0.0048)
M8. Spillover coefficient	0.191	0.192	(0.046)
M9. Elasticity of R&D investment to cost	−0.35	−0.35	(0.101)

Read:

- Table II is the key quantitative table. The headline estimate is $\rho_{\theta r} = 1.88$, which says that R&D is highly complementary to firm research productivity.
- The estimated persistence $\tilde{p} = 0.63$ implies that firms' private information matters dynamically, but distortions can still decay with age.
- The spillover estimate $\zeta = 0.018$ is small in absolute size but economically meaningful because it applies to the whole production environment.
- Table III shows that the model is calibrated to exactly the moments it needs: the patent-quality elasticity with respect to R&D, R&D intensity, sales growth, several dispersion moments in patent quality, a spillover coefficient, and an elasticity of R&D to user costs.
- The paper's goal is not perfect fit on every moment; it is to fit the moments that matter for the screening-versus-correction trade-off. In that sense the fit is quite good.

5.4 Table IV: goodness of fit on non-targeted moments

Read:

- Table IV shows that the model does reasonably well even away from its targeted moments.
- In the data, sales growth for the bottom 90% versus top 10% firms is 0.03 versus a model value of 0.04. R&D-to-sales for the bottom 90% is 0.038 in the data and 0.034 in the model; for the top 10%, 0.052 in the data and 0.042 in the model.
- The ratio of sales for old firms versus young firms is 1.73 in the data and 2.32 in the model. This is not perfect, but it suggests that the model captures life-cycle skewness in a broadly sensible way.

TABLE IV

GOODNESS OF FIT FOR NON-TARGETED MOMENTS.

	Data	Model
Sales growth bottom 90% vs. top 10%	0.03	0.04
R&D/Sales for bottom 90%	0.038	0.034
R&D/Sales for top 10%	0.052	0.042
Ratio sales for old firms vs. young firms	1.73	2.32

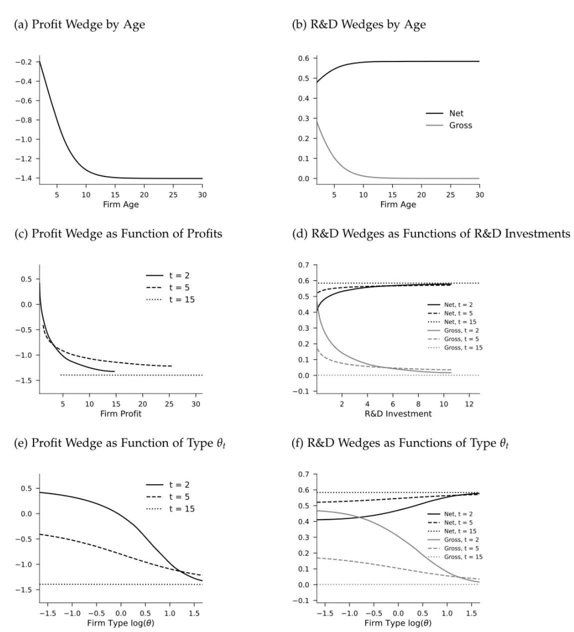

FIGURE 2.—Optimal profit and R&D wedges.

5.5 Figure 2: optimal profit and R&D wedges

Read:

- Panel A shows that the average profit wedge by age becomes more negative over the life cycle, moving from roughly −0.2 at very young ages to about −1.4 by age 15–20. In the paper's language, that means the profit side becomes more strongly subsidized relative to laissez-faire as screening pressure fades.
- Panel B shows that the *net* R&D wedge rises with age from about 0.48 to about 0.58, while the *gross* wedge declines toward zero. This is one of the most important pictures in the paper: part of the innovation incentive comes through the profit wedge, not just through explicit R&D subsidies.
- Panels C and E show that higher-profit and higher-type firms face lower profit wedges. Panels D and F show that they also receive larger *net* R&D incentives.
- The crucial interpretation is that the planner wants to reward high-productivity firms more, but it is better to do so through performance — lower profit wedges — than by handing them larger gross R&D subsidies.

5.6 Figure 3: optimal allocations

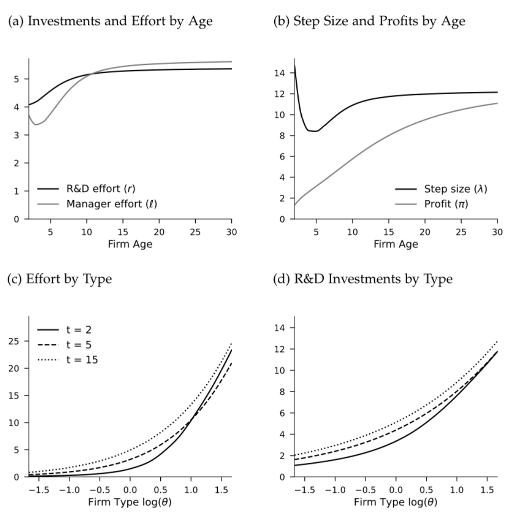

FIGURE 3.—Optimal allocations.

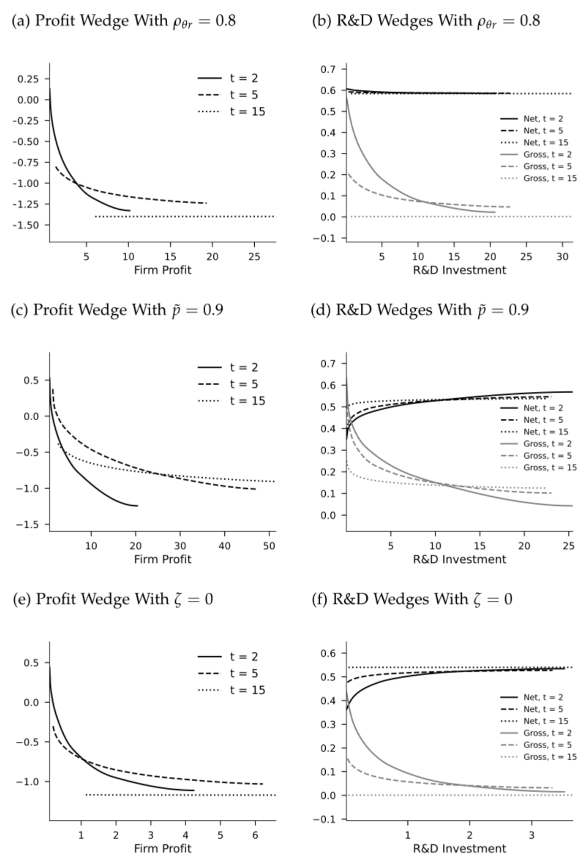

FIGURE 4.—Comparative statics: optimal profit and R&D wedges.

TABLE V

WELFARE FROM OPTIMAL SIMPLER POLICIES.

Policy Type		Welfare Achieved Relative to Full Optimum	
		Benchmark	No spillovers
A. Current US policy			
$T'(\pi) = 0.23$	$S'(M) = 0.19$	18%	31.1%
B. Optimal Linear			
$T'(\pi) = \tau_0$	$S'(M) = s_0$	89%	88.5%
C. Linear With Interaction Term			
$T'(\pi, M) = \tau_0 + \tau_1 M$	$S'(M) = s_0$	93.5%	93.7%
D. Heathcote–Storesletten–Violante (HSV)			
$T'(\pi) = \tau_0 - \tau_1 \pi^{\tau_2}$	$S'(M) = s_0 - s_1 M^{s_2}$	97.4%	98.2%
E. HSV Tax on Profits and Linear Subsidy			
$T'(\pi) = \tau_0$	$S'(M) = s_0 - s_1 M^{s_2}$	94.7%	95.6%
F. HSV Subsidy on R&D and Linear Profit Tax			
$T'(\pi) = \tau_0$	$S'(M) = s_0 - s_1 M^{s_2}$	97.3%	97.4%
G. HSV With Interaction Term			
$T'(\pi, M) = \tau_0 + \tau_3 M^{s_2} - \tau_1 \pi^{\tau_2}$	$S'(M) = s_0 - s_1 M^{s_2}$	97.4%	98.3 %

Note: The table shows the share of welfare from the full unrestricted optimum that is achieved by the optimal policy within each class. Each panel shows a different class. Column (1) shows the welfare relative to the benchmark optimum; Column (3) for the benchmark optimum but when there is no spillover ($\zeta = 0$).

Read:

- Panel A shows that both R&D investment and managerial effort rise with age, from around 4 to roughly 5.3–5.6 in the simulation.
- Panel B shows that the innovation step size starts high, dips early, and then recovers, while profits rise strongly over the life cycle.
- Panels C and D show the strongest monotonicity result in the paper: higher research productivity firms optimally choose much more effort and much more R&D investment.
- This figure gives the real allocation counterpart to Figure 2. Figure 2 is about wedges; Figure 3 is about what those wedges do.

5.7 Figure 4: comparative statics

Read:

- Panels A and B lower the complementarity parameter to $\rho_{\theta r} = 0.8 < \rho_{lr} = 1$. When this happens, the optimal R&D wedge rises, especially for low-productivity firms. The reason is that R&D is now less useful for mimicking high type.
- Panels C and D raise persistence to $\tilde{p} = 0.9$. The wedges then decay more slowly with age, because private information remains valuable for longer.
- Panels E and F set spillovers to zero. The wedges become smaller because the planner is now correcting only monopoly distortions, not externalities.
- This figure is the best summary of the paper's theory: complementarity, persistence, and spillovers are exactly the parameters that govern optimal policy.

5.8 Table V: welfare from simpler policies

Read:

- The current US-style policy — roughly a linear 23% corporate tax and a linear 19% R&D subsidy — achieves only **18%** of the welfare gain from the unrestricted optimum. Without spillovers, it achieves 31.1%.

- The best **linear** tax-plus-subsidy system already does surprisingly well: **89%** of the optimum.
- The best fully nonlinear HSV-type profit tax plus HSV-type R&D subsidy achieves **97.4%** of the optimum, and **98.2%** in the no-spillover comparison.
- The most informative comparison is between rows E and F. A *linear profit tax with a nonlinear R&D subsidy* achieves **97.3%** of the optimum, while a *nonlinear profit tax with a linear subsidy* achieves only **94.7%**.
- So the big practical lesson is: if you can make only one part of the policy nonlinear, make the **R&D subsidy** nonlinear.

6 Short summary

- **Method.** The paper studies optimal innovation policy as a dynamic mechanism design problem with spillovers, hidden firm types, and hidden R&D effort.
- **Core formula.** Optimal corporate and R&D wedges combine a Pigouvian spillover correction, a monopoly-distortion correction, and a screening term driven by asymmetric information.
- **Empirical finding.** In the data, observable R&D is highly complementary to private firm productivity ($\rho_{\theta r} = 1.88$), so broad R&D subsidies create informational rents if they are not designed carefully.
- **Policy implication.** The unrestricted optimum is complicated, but a simple policy with a linear profit tax and a nonlinear R&D subsidy performs almost as well.

7 Conclusion

7.1 Core conclusion

- The paper's main contribution is to bring asymmetric information into a dynamic firm-taxation model with spillovers.
- Once hidden research productivity and hidden effort are introduced, first-best Pigouvian logic is no longer enough. The planner must also screen firms and manage informational rents.
- Optimal policy therefore depends on a small set of economically interpretable parameters: complementarities, persistence, dispersion, and spillover strength.

7.2 Economic intuition

- If all firms were the same or perfectly observable, the government would mainly worry about correcting spillovers and monopoly distortions.
- Once firms privately know how good they are at turning R&D into innovation, subsidies do two things at once: they encourage effort, but they also attract mimicking by high-productivity firms.
- The planner's problem is therefore to decide when it is better to reward spending directly and when it is better to reward the profits that successful innovators generate.

- In the estimated model, performance-based incentives matter a lot: the planner wants lower profit wedges for more productive firms and only a carefully shaped nonlinear subsidy on R&D inputs.

7.3 Takeaway

This paper is important both methodologically and substantively. Methodologically, it extends dynamic mechanism design to a realistic firm problem with market power, hidden types, hidden effort, and spillovers. Substantively, it shows that the best innovation policy is not simply a large uniform subsidy. The central design problem is how to correct spillovers while screening firms whose innovative ability is privately known. In the data, a remarkably simple practical lesson emerges: keep the corporate tax fairly simple, and put the policy sophistication into the shape of the R&D subsidy.